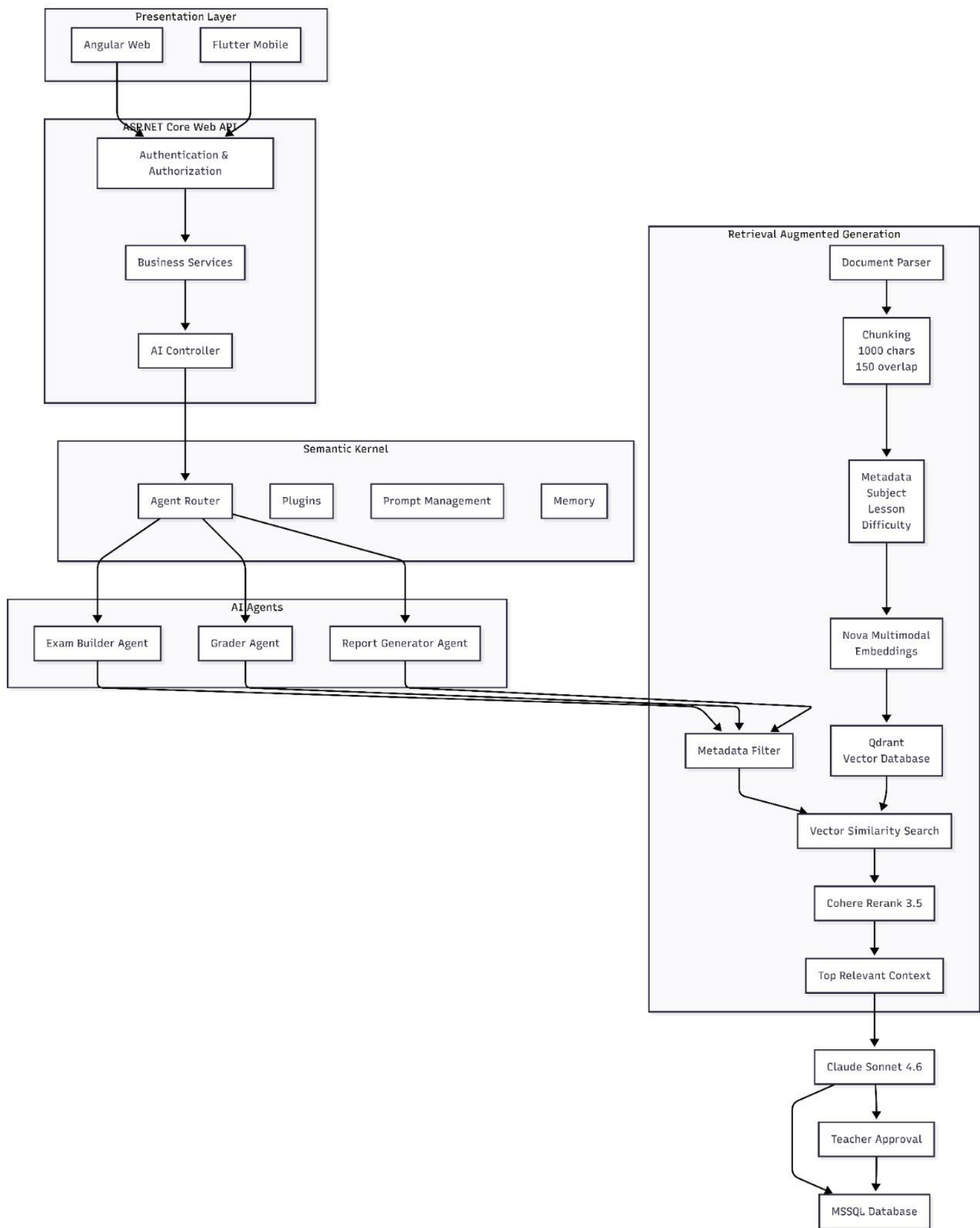
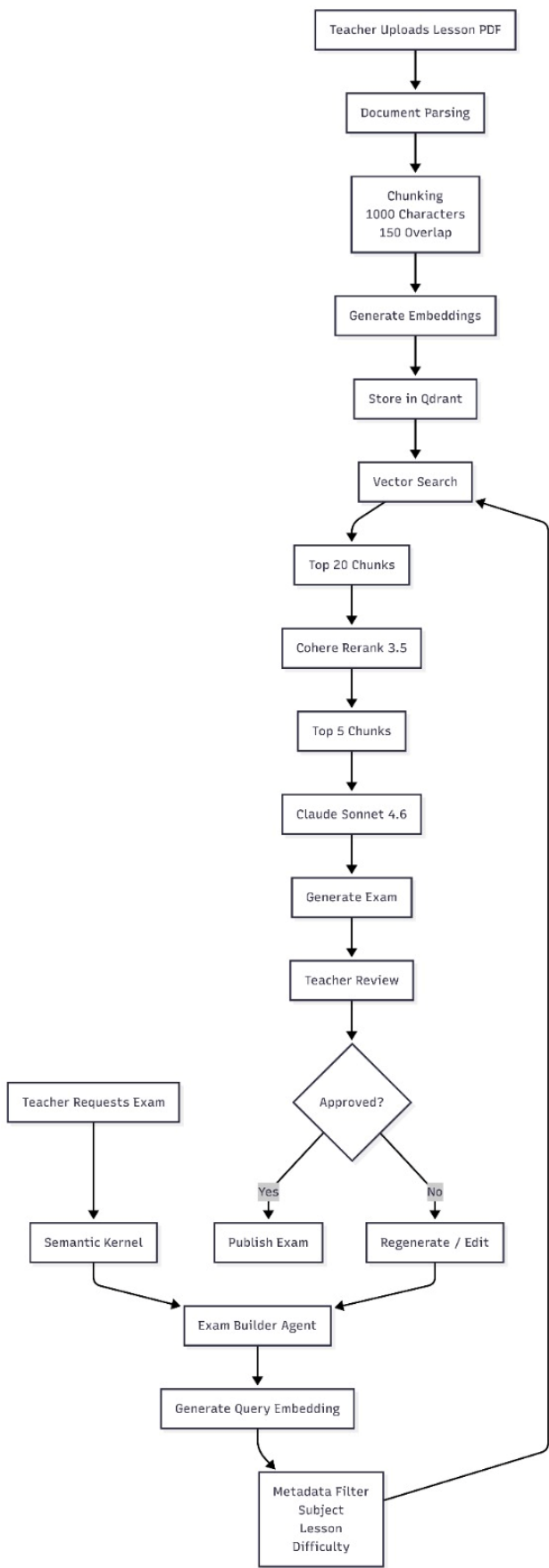


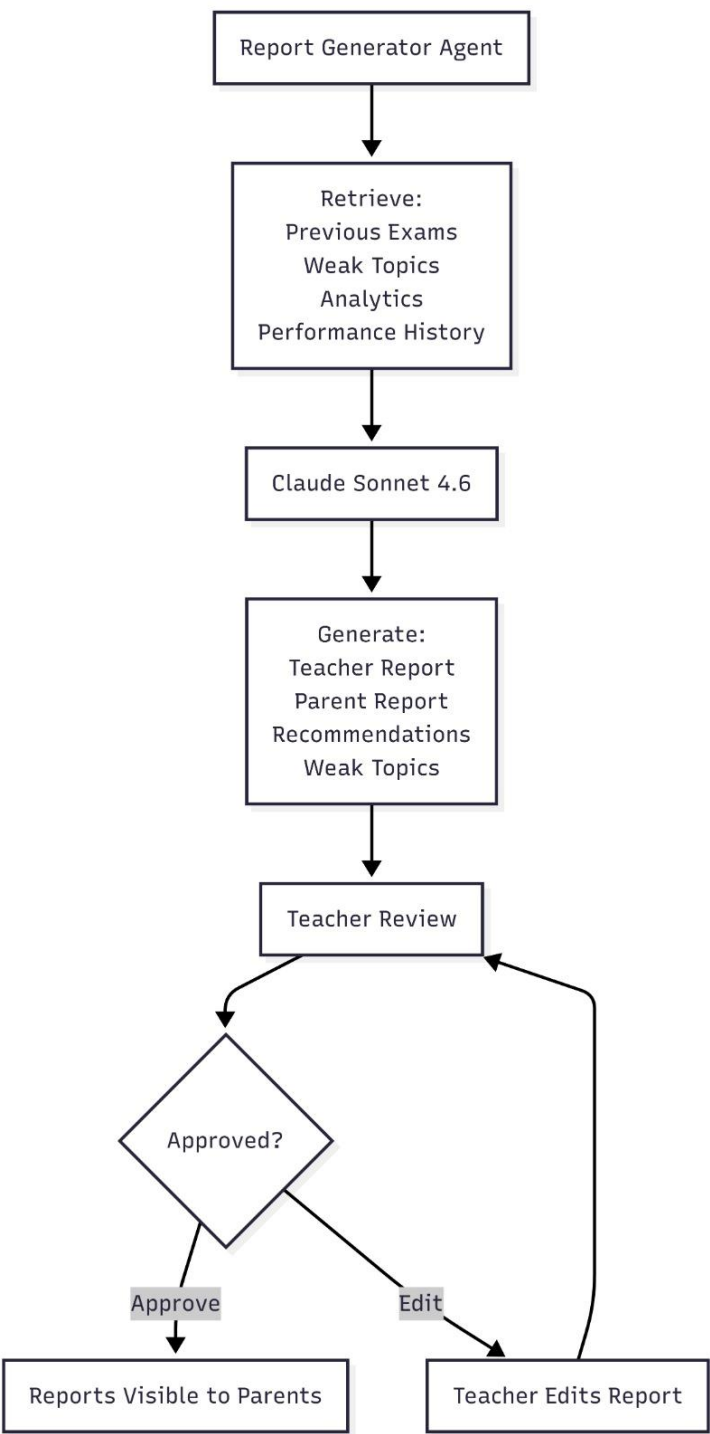
Overall AI Architecture:



Exam Generation Ai Flow:



Report Generator Ai Flow:



1. Document LLM Choice and Justification

LLM: Claude Sonnet 4.6.

Reasons: Excellent reasoning, strong Arabic support, reliable structured JSON generation, long context window, low hallucination when combined with RAG.

Agent Usage: Exam Builder generates exams, Grader evaluates answers, Report Generator produces analytics and recommendations.

Constraints: LLM never answers directly from its own knowledge; all generation is grounded using retrieved context.

2. **Agent framework:** Microsoft Semantic Kernel.

3. **Vector Database:** Qdrant Cloud.

4. **Embedding model:** Nova Multimodal embeddings.

5. Each agent responsibilities, inputs, and outputs:

Agent	Inputs	Processing	Outputs
Exam Builder	Lesson chunks, difficulty, exam type	Generates curriculum-based questions	Exam JSON
Grader	Student answers, marking scheme	Grades answers	Scores, feedback
Report Generator	Grades, analytics	Finds weak topics and recommendations	Reports

6. Chunking strategy:

Fixed-size chunking with overlap. Uploaded PDFs are parsed and split into chunks of 1,000 characters with a 150-character overlap. Each chunk is tagged with C# metadata classes containing SubjectId, LessonId, and DifficultyLevel to enable metadata pre-filtering inside Qdrant before running vector similarity matching.

7. Retrieval and Reranking Flow

Teacher query is converted to an embedding.

Metadata filters restrict search scope by subject, lesson and difficulty.

Qdrant performs vector similarity search.

Top results are reranked using Cohere Rerank 3.5.

Highest-ranked chunks are sent to Claude Sonnet 4.6.

Claude returns structured JSON (exam, grading or report) based only on retrieved context.

8. Human approval checkpoints & Hallucination guardrails:

- Teachers review generated exams before publishing, approve parent reports before visibility, and may edit questions, reports or override AI grading.

- Model temperature is locked to 0.0 for grading and exam generation, and Semantic Kernel system prompts enforce a strict guardrail: "Answer ONLY using the provided textbook context. If the context does not contain the information, reply with 'DATA_UNAVAILABLE.' Do not invent details."